

The role of Gliadel wafers in the treatment of high-grade gliomas.

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Glioblastoma multiforme (GBM) is the most aggressive brain tumor. Standard treatment includes surgery, radiation and chemotherapy. Prognosis is dismal with an average survival of approximately 1 year. Gliadel wafers are one treatment option, working as a source for local chemotherapy delivery. Their use is controversial with questionable survival benefit and potential side effects. We reviewed the literature in an effort to clarify their role in the treatment of high-grade gliomas. A systematic PubMed search was performed using the keywords 'Gliadel', 'carmustine' or 'BCNU wafers' in newly diagnosed high-grade glioma patients. Treatment regimen, and median survival were analyzed. Adverse event ratio was calculated by computing the number of adverse events in a study per patient receiving carmustine wafers. Nineteen studies with 795 patients were included in our review. Survival was 8.7-22.6 months with a mean overall survival (OS) of 16.2 months (control survival is approximately 14 months with surgery and adjuvant chemoradiotherapy). Adverse event ratio using Gliadel wafers in control group. Complication rate was 42.7%. Gliadel wafers may marginally increase survival and local control in newly diagnosed GBM patients but are associated with a high complication rate; therefore, we do not recommend using Gliadel wafers in patients with GBM. Further research may be warranted once a safer alternative to Gliadel wafers has been introduced.

Bevacizumab for the treatment of glioblastoma.

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Glioblastoma (GBM) or grade IV glioma is the most common primary brain tumor in adults. Standard treatment median overall survival (OS) is only 14-15 months and less than 10% of patients will survive 5 years after diagnosis. There is no standard treatment in recurrent GBM and OS ranges from 3 to 9 months. GBM is 1 of the most vascularized human tumors and GBM cells produce vascular endothelial growth factor (VEGF). Bevacizumab, a humanized monoclonal antibody against VEGF, has demonstrated activity in vitro and in phase II trials in relapse, as well as in 1 phase III trial as first line therapy. Bevacizumab also improves quality of life for patients suffering GBM. This paper reviews the mechanism of action of bevacizumab, its metabolism and pharmacokinetic profile. It summarizes the clinical studies in recurrent and newly diagnosed GBM, its potential side effects and complications and its place in therapy.

Editorial: on the road to multi-modal and pluri-disciplinary treatment of glioblastomas.

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Despite major advances in the management of malignant gliomas of which glioblastomas represent the ultimate grade of malignancy, they remain incurable. Indeed, glioblastoma patients have a median survival expectancy of only 14 months on the current standard treatment of surgical resection to the extent which is feasible, followed by adjuvant radiotherapy plus temozolomide given concomitantly with and after radiotherapy (Lefranc et al., J Clin Oncol 23:2411-2422, 2005; Expert Rev Anticancer Ther 6:719-732, 2006; Stummer et al., Neurosurgery 62:564-576, 2008). Accordingly, the present editorial discusses (1) the high cell motility and resistance to apoptosis which characterise glioblastoma growth

and malignancy with respect to the failure of conventional therapy, (2) ways to overcome apoptosis resistance and the real hope offered by temozolomide, (3) targeted chemotherapeutic approaches and the disappointing results obtained in monotherapy but their potential in combination therapy, (4) anti-migratory strategies that could supplement conventional therapy notably by inhibiting a new target; the $\alpha 1$ subunit of the sodium pump, (5) dendritic cell therapy, (6) cancer stem cell targeting and finally (7) topical therapies and new surgical approaches for more radical resection which could be used to complement multi-modal treatments within a multi-disciplinary approach.

New agents targeting angiogenesis in glioblastoma.

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Glioblastoma is the most common malignant glioma in adults, and despite recent advances in standard treatment, the prognosis still remains dismal, with a median survival of 15 months. The incorporation of bevacizumab in the standard treatment of relapsed glioblastoma has been a significant step towards combining targeted agents with chemotherapy, and there is an increasing number of new antiangiogenic agents in various stages of development, that are being tested both in relapsed and newly diagnosed disease, alone or in combination with standard treatment. The relatively favorable toxicity profile for most of them presents an advantage, but several concerns arise regarding their actual efficacy on the clinical level and the most efficient schedule of administration for each of them, as their molecular targets and patterns of action may vary significantly. This may lead to future modifications of the current rational of administering these agents concomitantly with initial chemotherapy or maintenance treatment.

Ketogenic metabolic therapy in conjunction with standard treatment for glioblastoma: A case report.

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Glioblastoma (GBM) is the most common primary malignant brain tumour in adults. The standard of care consists of surgical resection and concurrent chemoradiation, followed by adjuvant temozolomide chemotherapy. This protocol is associated with a median survival of 12-15 months, and 3 years. Ketogenic metabolic therapy (KMT) targets cancer cell metabolism by restricting glucose availability and evoking differential stress resistance and sensitization, which may augment the standard treatments and lead to therapeutic benefit. The present study reports the case of a 64-year-old woman with isocitrate dehydrogenase (IDH)-wildtype GBM who pursued the standard treatment protocol in conjunction with an intensive, multimodal KMT program for 3 years. The KMT program consisted of a series of prolonged (7-day, fluid-only) fasts, which were specifically timed to maximize the tolerability and efficacy of the standard treatments, combined with a time-restricted ketogenic diet on all other days. During the first and second treatment years the patient sustained a glucose ketone index (GKI) of 1.65 and 2.02, respectively, which coincided with complete clinical improvement, a healthy body-mass index and a high quality of life, with no visible progressive tumour detected on imaging at the end of the second year. In the setting of the death of an immediate family member leading to increased life stress, slightly relaxed KMT adherence, and a higher GKI of 3.20, slow cancer progression occurred during the third year. The adverse effects attributed to KMT were mild. Despite the limitations of this case report, it highlights the

feasibility of implementing the standard treatment protocol for GBM in conjunction with an intensive, long-term, multimodal and specifically timed KMT program, the potential therapeutic efficacy of which may depend upon achieving as low a GKI as possible.

Rindopepimut: a promising immunotherapeutic for the treatment of glioblastoma multiforme.

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Glioblastoma multiforme (GBM) is the most common and aggressive glial cell-derived primary tumor. Current standard of care for patients with GBM includes maximal tumor resection plus adjuvant radiotherapy and temozolomide chemotherapy, increasing median overall survival to a mere 15 months from diagnosis. Because these therapies are inherently nonspecific, there is an increased likelihood of off-target and incomplete effects; therefore, targeted modalities are required for enhanced safety and efficacy. Rindopepimut is emerging as a safe and potentially effective drug for the treatment of GBM. Rindopepimut consists of a 14-mer peptide that spans the length of EGF receptor variant III, a mutant variant of EGF receptor found on approximately 30% of primary GBM, conjugated to the carrier protein keyhole limpet hemocyanin. Vaccination with rindopepimut has been shown to specifically eliminate cells expressing EGF receptor variant III. Phase II clinical trials have suggested that vaccination of newly diagnosed GBM patients with rindopepimut plus adjuvant granulocyte-macrophage colony-stimulating factor results in prolonged progression-free and overall survival with minimal toxicity. This review will outline the development of rindopepimut, as well as the current status of this vaccine.

Identification of a cullin5-RING E3 ligase transcriptome signature in glioblastoma multiforme.

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Glioblastoma multiforme (GBM) is the deadliest type of brain tumor. The median survival time for patients with GBM is only 15 months, even following maximal surgical resection and chemotherapy and radiation therapy. A genetic biomarker could enable a paradigm shift in precise diagnosis, personalized therapeutics and prognosis. In this study, we employed the Chinese Glioma Genome Atlas, The Cancer Genome Atlas, and the Ivy Glioblastoma Atlas Project databases for RNA sequencing (RNA-seq) analysis and clinicopathological studies. We demonstrated that elevated expression of the RNF7, TCEB1, SOCS1 and SOCS3 genes, which encode components of cullin5-RING E3 ligase (CRL5), predict unfavorable GBM prognoses. In GBM and glioma cases carrying IDH1 mutations, SOCS1 and SOCS3 methylation was increased and their expression was downregulated. This study has thus identified a simple transcriptome signature for GBM prognosis.

Temozolomide and other potential agents for the treatment of glioblastoma multiforme.

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This article provides historical and recent perspectives related to the use of temozolomide for the

treatment of glioblastoma multiforme. Temozolomide has quickly become part of the standard of care for the modern treatment of stage IV glioblastoma multiforme since its approval in 2005. Yet despite its improvements from previous therapies, median survival remains approximately 15 months, with a 2-year survival rate of 8% to 26%. The mechanism of action of this chemotherapeutic agent, conferred advantages and limitations, treatment resistance and rescue, and potential targets of future research are discussed.

Nanotechnology for Treatment of Glioblastoma Multiforme.

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Glioblastoma multiforme (GBM), a grade IV astrocytoma as defined by the World Health Organization (WHO) criteria, is the most common primary central nervous system tumor in adults. After treatment with the current standard of care consisting of surgical resection, concurrent temozolomide (TMZ), and radiation, the median survival is only 15 months. The limited and less-effective treatment options for these highly aggressive GBMs call for the development of new techniques and the improvement of existing technologies. Nanotechnology has shown promise in treating this disease, and some nanomaterials have demonstrated the ability to cross the blood-brain barrier (BBB) and remain in GBM tissues. Although the retention of nanoparticles (NPs) in GBM tissue is necessary to elicit an antitumor response, the delivery of the NP needs to be enhanced. Current research in nanotechnology is directed at increasing the active targeting of GBM tissue not only for the aid of chemotherapeutic drug delivery but also for imaging studies. This review is aimed at describing advancements in increasing nanotechnology specificity to GBM tissue.

New therapeutic approaches for malignant glioma: in search of the Rosetta stone.

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Malignant gliomas are heterogeneous, diffuse and highly infiltrating by nature. Despite wide surgical resection and improvements in radio- and chemotherapies, the prognosis of patients with glioblastoma multiforme remains extremely poor, with a median survival time of only 14.5 months from diagnosis to death. Particular challenges for glioblastoma multiforme therapy are posed by limitations in the extent of feasible surgical resections, distinct tumor heterogeneity, difficulties in drug delivery across the blood-brain barrier and low drug distribution within the tumor. Therefore, new paradigms permitting tumor-specific targeting and extensive intratumoral distribution must be developed to allow an efficient therapeutic delivery. This review highlights the latest advances in the treatment of glioblastoma multiforme and the recent developments that have resulted from the interchange between preclinical and clinical efforts. We also summarize and discuss novel therapies for malignant glioma, focusing on advances in the following main topics of glioblastoma multiforme therapy: immunotherapy, gene therapy, stem cell-based therapies and nanotechnology. We discuss strategies and outcomes of emerging therapeutic approaches in these fields, and the main challenges associated with the integration of discoveries that occur in the laboratory into clinical practice.